

# PAPER Breeding-Blanket and Neutron-Multiplier Concept Down-Select for a Compact Spherical-Tokamak Breeder: Ceramic versus Liquid-Metal Options Walking the TBR Ladder to Self-Sufficiency

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## Abstract

For a breeder whose mission is tritium, helium-3 and 14 MeV neutrons—never net electricity—the blanket is the product line, and its design reduces to a concept down-select on one axis: the net tritium breeding ratio (TBR). We state the coupled neutron-transport, fuel-cycle, conjugate-heat-transfer and permeation problems formally, then rank the candidate blanket and neutron-multiplier concepts for the Hyperion breeder (design point  $Q = 3.076$ ,  $P_{\text{fus}} = 85.04$  MW,  $I_p = 9.66$  MA,  $\delta = -0.30$ ,  $B_0 = 8$  T,  $\kappa = 2.0$ ,  $R_0 = 1.20$  m, aspect ratio  $A = 2.5$ , centrepost peak field 16.84 T) on a common three-dimensional continuous-energy Monte-Carlo footing (OpenMC/ENDF-B-VIII.0,  $1.2 \times 10^8$  histories) and against the self-sufficiency bar  $\Lambda \geq 1.05$ . A bare recipe with no dedicated multiplier breeds only 0.742 and depletes a 2 kg inventory in 1.49 fpy; a beryllium-packed ceramic breeder (Be 9.6–11.8 vol %) reaches net TBR **1.026–1.062** (inboard mantle 0.2299 + outboard blanket 0.7960, leakage 0.12064), and a dedicated 5 cm solid-beryllium multiplier layer lifts the same lever to **1.18–1.20**; outboard-blanket thickness alone moves it to 1.24–1.25 at 65–80 cm. Self-sufficiency is *resolved* in the as-drawn three-dimensional CAD geometry at net TBR 1.12 (single null) and 1.07 (double null). On the breeder-form axis, a self-cooled liquid-lithium blanket passes thermally with a reduced-activation ferritic-martensitic (RAFM) structure at 516 °C (+34 margin) where an as-configured helium-cooled ceramic (HCPB) bed runs to 712 °C, and liquid lithium retains tritium roughly 6.7 orders of magnitude better than tin. The selection is a beryllium-multiplied ceramic blanket as the baseline breeder with a self-cooled-lithium blanket as the thermally robust alternative; the multiplier thickness is the one lever that trades against the centrepost shield in the radial budget.

**Keywords:** tritium breeding ratio; beryllium multiplier; ceramic breeder; liquid lithium; blanket down-select; neutronics; conjugate heat transfer

## 1 Introduction

### 1.1 The breeder as a materials machine

The Hyperion breeder is a materials machine, not a power plant. Its outputs are tritium for the fuel cycle, helium-3 banked from tritium decay, and a 14 MeV neutron source for isotope production and irradiation testing; its success criterion is breeding, never net electricity [1, 2, 3]. That reframing is not rhetorical. It moves the blanket from a piece of balance-of-plant into the central product, and it makes the entire engineering question a concept down-select on a single figure of merit—the net tritium breeding ratio  $\Lambda$  (tritons bred per fusion neutron, after leakage and the inboard shield). Every candidate blanket and every candidate neutron multiplier is admitted or rejected by where it sits on that one axis relative to the self-sufficiency bar, and every other property—thermal margin, tritium retention, activation, maturity—is a constraint that a passing concept must additionally clear.

### 1.2 Why a compact spherical tokamak makes breeding hard

A compact spherical tokamak (ST) confines a large plasma current in a small volume, which is what makes it an attractive neutron source and breeder platform [32, 33], but it also makes the

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breeding problem harder than in a large aspect-ratio device. The inboard side is occupied by the centrepost carrying the toroidal field (16.84 T peak at the conductor) and by the tungsten-carbide shield that protects it; there is almost no room for an inboard breeding mantle. The consequence is geometric and unforgiving: most of the tritium must be bred in the outboard blanket, the solid angle available for breeding is reduced, and the neutron budget is tight. In our reference three-dimensional model the inboard mantle contributes only  $\Lambda_{\text{in}} = 0.2299$  against the outboard  $\Lambda_{\text{out}} = 0.7960$  (§4). Neutron *multiplication* is therefore not an optimization luxury; it is the enabling lever that carries a compact ST from sub-unity breeding to self-sufficiency.

### 1.3 Prior art

The two mature solid-breeder lines are the helium-cooled pebble bed (HCPB), in which lithium-orthosilicate or metatitanate ceramic pebbles are interleaved with a beryllium or beryllide neutron multiplier [19, 20, 21], and the helium-cooled or water-cooled lithium-lead (HCLL/WCLL) lines, in which the Pb-15.7Li eutectic is both breeder and multiplier [22, 21]. Liquid-metal concepts—self-cooled or dual-cooled lithium and lithium-lead—trade the pebble bed’s manufacturing maturity for a flowing breeder that carries its own tritium out of the vessel and tolerates higher volumetric heating [23, 24, 25]. Neutron multiplication in fusion blankets has been understood since the earliest blanket studies [17, 18]; the modern beryllide multipliers ( $\text{Be}_{12}\text{Ti}$ ,  $\text{Be}_{12}\text{V}$ ) were developed precisely to raise the multiplier operating temperature and suppress the helium-driven swelling that limits metallic beryllium [26, 27, 28]. The condition for tritium self-sufficiency—that the net TBR exceed unity by a margin sufficient to cover losses, radioactive decay, startup inventory and a doubling requirement—was placed on a rigorous footing by Abdou and co-workers [29, 30, 31], whose analysis frames the design target we adopt here.

### 1.4 Contribution

This paper puts the candidate blanket and multiplier concepts side by side on one common, physically explicit footing and states the selection. Our specific contributions are: (i) a formal statement of the coupled neutron-transport, fuel-cycle, conjugate-heat-transfer and tritium-permeation problems that jointly define blanket viability (§2); (ii) a common three-dimensional continuous-energy Monte-Carlo evaluation of every concept on the frozen operating point, executed on the NVIDIA GPU HPC campaign (§3); (iii) a decomposition of the net TBR into an additive inboard/outboard ledger and into separable design levers—multiplication, thickness, coverage—so that the down-select is transparent rather than a single opaque number (§4); and (iv) the selection itself, a beryllium-multiplied ceramic baseline with a self-cooled-lithium alternative, together with the one radial-budget trade that governs it (§5–§7). Throughout we treat  $\Lambda$  as a design *lever*—a family of valid blanket points—rather than a single fixed number, because the compact ST’s shield-dominated inboard makes coverage, not intrinsic breeding chemistry, the binding constraint. The result complements the companion Kronos studies that established the cross-library breeding-fleet case [4], the CAD-faithful three-dimensional neutronics that resolves the net TBR in real geometry [5], the dedicated beryllium and  $\text{Be}_{12}\text{Ti}$  multiplier down-select [7], the per-unit tritium ladder and doubling time [8], the blanket and divertor thermal-hydraulics [11], and the liquid-lithium-over-tin wall-chemistry choice [12].

## 2 Governing equations and the formal problem

The down-select is decided by four coupled physics problems. We state each as a boundary- or initial-value problem, identify the functional that maps a blanket design vector to its figure of merit, and then cast the selection itself as a constrained decision.

### 2.1 The design vector and the selection problem

Let a blanket concept be described by a design vector

$$\mathbf{d} = (m, f_{\text{Be}}, t_{\text{b}}, e_6, f_{\text{cov}}, \phi_{\text{form}}), \quad (1)$$

with multiplier type  $m \in \{\emptyset, \text{packed Be}, \text{solid Be}, \text{Be}_{12}\text{Ti}\}$ , beryllium volume fraction  $f_{\text{Be}}$ , outboard breeding-zone radial thickness  $t_{\text{b}}$ , lithium-6 enrichment  $e_6$ , effective three-dimensional breeding coverage  $f_{\text{cov}}$ , and breeder form  $\phi_{\text{form}} \in \{\text{ceramic}, \text{liquid Li}, \text{PbLi}\}$ . Three scalar responses are computed from  $\mathbf{d}$  at the frozen operating point: the net breeding ratio  $\Lambda(\mathbf{d})$ , the peak structure temperature  $T_{\text{max}}(\mathbf{d})$ , and the tritium retention/permeation margin  $\Pi(\mathbf{d})$ . The down-select is the

constrained feasibility-then-ranking problem

$$\text{find } \mathbf{d} \in \mathcal{D} \quad \text{such that} \quad (S1) \quad \Lambda(\mathbf{d}) \geq \Lambda^* = 1.05, \quad (2)$$

$$(S2) \quad T_{\max}(\mathbf{d}) \leq T_{\lim} = 550^\circ\text{C}, \quad (3)$$

$$(S3) \quad \text{waste class}(\mathbf{d}) \preceq \text{Class C}, \quad (4)$$

$$(S4) \quad t_b(\mathbf{d}) + t_{\text{sh}} \leq R_{\text{budget}} \quad (\text{radial-space closure}), \quad (5)$$

ranked by technological maturity and fuel-cycle robustness among the feasible set.

Equations (2)–(5) are the machine-checkable contract that the rest of this section makes concrete.

## 2.2 Tritium fuel-cycle balance and the self-sufficiency condition

The mobile in-vessel tritium inventory  $I(t)$  obeys the fuel-cycle balance

$$\frac{dI}{dt} = \dot{N}_{\text{bred}} - \dot{N}_{\text{burn}} - \lambda_T I - \dot{L}_{\text{loss}}, \quad \dot{N}_{\text{bred}} = \Lambda \dot{N}_{\text{burn}}, \quad (6)$$

where  $\dot{N}_{\text{burn}} = \dot{m}_{\text{burn}}$  is the tritium burn rate,  $\lambda_T = \ln 2 / \tau_{1/2}$  is the tritium decay constant ( $\tau_{1/2} = 12.32 \text{ yr}$ ), and  $\dot{L}_{\text{loss}}$  collects permeation, trapping and processing losses. The burn rate follows directly from the fusion power. With  $E_{\text{DT}} = 17.6 \text{ MeV}$  per D–T reaction, each reaction consuming one triton,

$$\dot{m}_{\text{burn}} = \frac{P_{\text{fus}}}{E_{\text{DT}}} m_T = \frac{85.04 \text{ MW}}{17.6 \text{ MeV}} (3.016 \text{ u}) \approx 3.02 \times 10^{19} \text{ s}^{-1} \approx 4.769 \text{ kg fpy}^{-1}, \quad (7)$$

which is the register anchor (BR-L1-A6). Writing the losses as an effective fraction folded into the net ratio, the quasi-steady tritium surplus available as product is the affine law

$$S(\Lambda) = \dot{m}_{\text{burn}} (\Lambda - 1) - \lambda_T I_{\text{ss}}, \quad (8)$$

whose slope is exactly  $\dot{m}_{\text{burn}}$ . Neglecting the small decay term against the anchor calibration,  $S = \dot{m}_{\text{burn}} (\Lambda - 1)$  reproduces the register outputs to the reported precision:  $S = 1.6215 \text{ kg fpy}^{-1}$  at  $\Lambda = 1.34$  and  $S = 3.8152 \text{ kg fpy}^{-1}$  at  $\Lambda = 1.80$  (BR-OUT-1), a per-unit export of  $1 \text{ kg fpy}^{-1}$  at  $\Lambda = 1.2097$  and  $2 \text{ kg fpy}^{-1}$  at  $\Lambda = 1.4194$  (BR-L2-A11h). The self-sufficiency condition (S1) is the requirement that (6) admit a non-decreasing inventory once startup, decay and a doubling allowance are covered,

$$\Lambda \geq \Lambda^* = 1 + \varepsilon_{\text{loss}} + \frac{\lambda_T I_{\text{ss}}}{\dot{m}_{\text{burn}}} + \varepsilon_{\text{doubling}} \approx 1.05, \quad (9)$$

and the startup inventory required before breeding begins is  $I_{\text{start}} = \dot{m}_{\text{burn}} \tau_{\text{start}}$ , i.e.

$2.38/4.77/9.54 \text{ kg}$  for  $\tau_{\text{start}} = 0.5/1/2 \text{ yr}$  (BR-L1-A6). Equation (6) also makes the failure mode quantitative: at the coverage-honest  $\Lambda = 0.742$  the source term is negative,  $S \simeq -1.28 \text{ kg fpy}^{-1}$ , and a  $2 \text{ kg}$  inventory falls to  $0.64 \text{ kg}$  at one year and depletes at  $1.49 \text{ fpy}$ . Sub-unity breeding is therefore not a small deficit to be trimmed; it is a hard no-go, and multiplication is the only lever that closes it.

## 2.3 The neutron-transport problem and the TBR functional

The angular neutron flux  $\psi(\mathbf{r}, \hat{\Omega}, E)$  in the blanket obeys the steady linear Boltzmann transport equation

$$\hat{\Omega} \cdot \nabla \psi + \Sigma_t(\mathbf{r}, E) \psi = \int_{4\pi} d\hat{\Omega}' \int_0^\infty dE' \Sigma_s(\mathbf{r}, E' \rightarrow E, \hat{\Omega}' \rightarrow \hat{\Omega}) \psi(\mathbf{r}, \hat{\Omega}', E') + \frac{S_n(\mathbf{r}, E)}{4\pi}, \quad (10)$$

with  $\Sigma_t, \Sigma_s$  the macroscopic total and double-differential scattering cross sections and  $S_n$  the fusion neutron source. The source is not free: its spatial shape is the plasma neutron emissivity, set by the free-boundary MHD equilibrium poloidal flux  $\psi_{\text{eq}}(R, Z)$  through the fusion reactivity  $S_n(\mathbf{r}) \propto \frac{1}{4} n_D n_T \langle \sigma v \rangle_{\text{DT}}(T_i(\mathbf{r}))$ , emitting a  $14.06 \text{ MeV}$  spectrum. The scalar flux is  $\phi(\mathbf{r}, E) = \int_{4\pi} \psi d\hat{\Omega}$ . The gross tritium breeding ratio is the reaction-rate functional

$$\Lambda_{\text{gross}} = \frac{1}{N_{\text{src}}} \int_V d\mathbf{r} \int_0^\infty dE \left[ N_6 \sigma_{(n,t)}^{\text{Li6}}(E) + N_7 \sigma_{(n,n'\alpha)t}^{\text{Li7}}(E) \right] \phi(\mathbf{r}, E), \quad (11)$$

summing the exothermic  ${}^6\text{Li}(n, \alpha)t$  channel (dominant at thermal and epithermal energies) and the endothermic  ${}^7\text{Li}(n, n'\alpha)t$  channel (a fast-spectrum contributor above  $\sim 2.5$  MeV). The *net* ratio subtracts what the geometry loses,

$$\Lambda = \Lambda_{\text{gross}} - L_{\text{leak}} - P_{\text{par}}, \quad L_{\text{leak}} = \frac{1}{N_{\text{src}}} \oint_{\partial V} \mathbf{J} \cdot \hat{\mathbf{n}} \, dA, \quad (12)$$

with  $L_{\text{leak}}$  the boundary leakage and  $P_{\text{par}}$  the parasitic capture on structure, coolant and shield. Neutron multiplication enters as a source-doubling reaction that raises the population available to (11): per source neutron the available population is  $1 + M$ , with

$$M = \frac{1}{N_{\text{src}}} \int_V d\mathbf{r} \int dE \left[ N_{\text{Be}} \sigma_{(n,2n)}^{\text{Be}}(E) + 2 N_{\text{Be}} \sigma_{(n,3n)}^{\text{Be}}(E) + \dots \right] \phi. \quad (13)$$

Because  $\sigma_{(n,2n)}$  for  ${}^9\text{Be}$  has a threshold near 1.85 MeV and rises through the 14 MeV source region, multiplication is most effective on the un-degraded fast flux at the front of the blanket—which is why a dedicated multiplier layer placed *ahead* of the breeder outperforms the same beryllium distributed through a moderated bed. Finally, the shield-dominated inboard makes the net ratio naturally additive over the poloidal solid angle,

$$\Lambda = \Lambda_{\text{in}} + \Lambda_{\text{out}}, \quad (14)$$

the decomposition we report and exploit in §4.

#### 2.4 Levers and the coverage discount

Equations (11)–(13) expose  $\Lambda$  as a smooth functional of  $\mathbf{d}$  with three separable, monotone levers. The *multiplication* lever raises the population feeding (11); at fixed geometry  $\partial\Lambda/\partial f_{\text{Be}} > 0$  until self-shielding of the beryllium saturates  $M$ . The *thickness* lever raises the optical depth of the breeding zone; for an exponentially attenuating fast flux the incremental capture saturates as

$$\Lambda(t_b) \simeq \Lambda_{\infty} - (\Lambda_{\infty} - \Lambda_0) e^{-t_b/\ell}, \quad (15)$$

with  $\ell$  an effective attenuation length, so  $\partial\Lambda/\partial t_b > 0$  but diminishing. The *coverage* lever is multiplicative and is the compact ST's binding term: the reduced-order chemistry is discounted by the fraction of the neutron-emitting solid angle actually subtended by breeding material,

$$\Lambda^{3D} \approx f_{\text{cov}} \Lambda^{1D}, \quad 0 < f_{\text{cov}} \leq 1. \quad (16)$$

A literal one-dimensional recipe that breeds above unity can fall below it once the inboard shield, ports and divertor cut-outs are resolved; conversely a modest chemistry margin plus high coverage clears the bar. This is why the three-dimensional evaluation (§3) is not a refinement of the ladder but its arbiter.

#### 2.5 Breeder-form viability: conjugate heat transfer

A concept that breeds must also be coolable. In the solid structure and breeder the temperature field obeys the steady conduction equation with the volumetric nuclear heating  $q_n'''(\mathbf{r})$  taken directly from the neutronics deposition tally,

$$\nabla \cdot (k(T) \nabla T) + q_n'''(\mathbf{r}) = 0, \quad (17)$$

coupled at the wetted surfaces to the coolant energy equation  $\rho c_p (\partial_t T + \mathbf{u} \cdot \nabla T) = \nabla \cdot (k \nabla T) + \Phi$  closed by a turbulent wall-heat-transfer coefficient  $h = \text{Nu} k_f / D_h$  with  $\text{Nu} = \text{Nu}(\text{Re}, \text{Pr})$ . The coolant mass flow required to remove blanket power  $Q_{\text{blk}}$  at allowable temperature rise  $\Delta T$  is  $\dot{m} = Q_{\text{blk}} / (c_p \Delta T)$ ; for the helium loop ( $c_p = 5193 \text{ J kg}^{-1} \text{ K}^{-1}$ ,  $\Delta T = 200 \text{ K}$ ,  $Q_{\text{blk}} = 95 \text{ MW}$ ) this is  $\dot{m} = 91.7 \text{ kg s}^{-1}$  (BV-06). The viability constraint (S2) is that the peak structure temperature stay under the RA FM limit,

$$T_{\text{max}} = \max_{\mathbf{r} \in \text{struct}} T(\mathbf{r}) \leq T_{\text{lim}} = 550^\circ \text{C}. \quad (18)$$

For the helium-cooled ceramic (HCPB) bed this couples back to geometry: (18) is met only if the first-wall channel is held to  $\leq 6 \text{ mm}$  depth and  $\leq 15 \text{ mm}$  pitch (BR-SX-15); a self-cooled liquid-lithium blanket carries the heat in the flowing breeder itself and meets (18) with margin.

### 2.6 Tritium transport in the breeder: diffusion-trapping and permeation

The fuel-cycle loss term  $\dot{L}_{\text{loss}}$  in (6) and the retention margin  $\Pi(\mathbf{d})$  are set by tritium transport through the breeder and structure. We model the mobile concentration  $C_m$  and trapped concentration  $C_t$  by the McNabb–Foster diffusion-trapping system

$$\frac{\partial C_m}{\partial t} = \nabla \cdot (D(T) \nabla C_m) - \frac{\partial C_t}{\partial t} + S_g, \quad (19)$$

$$\frac{\partial C_t}{\partial t} = k_{\text{tr}}(T) C_m (n_t - C_t) - p_{\text{dt}}(T) C_t, \quad (20)$$

with  $D(T) = D_0 e^{-E_D/k_B T}$  the diffusivity,  $k_{\text{tr}}, p_{\text{dt}}$  the Arrhenius trapping/de-trapping rates,  $n_t$  the trap density and  $S_g$  the generation rate. At a wetted surface the boundary condition is the Sieverts law  $C_m = K_S(T) \sqrt{p_{T_2}}$ , and a permeation-barrier coating of permeation-reduction factor (PRF) reduces the through-wall steady flux by the same factor,  $J_{\text{perm}} = \Phi_{\text{perm}} \sqrt{p_{T_2}} / (\text{PRF} \cdot L)$ . The retention margin  $\Pi$  that differentiates breeder forms is dominated by the breeder’s tritium solubility: pure lithium retains tritium roughly 6.7 orders of magnitude more strongly than tin and 4.6 orders more strongly than Pb-15.7Li, giving a  $\sim 7000$  h breakthrough buffer (BR-SX-03), while candidate barrier coatings ladder as  $\text{Al}_2\text{O}_3$  ( $\text{PRF} \sim 10^3$ )  $>$   $\text{Er}_2\text{O}_3$  ( $\sim 3 \times 10^2$ )  $>$   $\text{TiC} > \text{TiN}$ . Equations (19)–(20) therefore close the mission-critical accountancy: the breeder that retains tritium best protects the very inventory that (6) tracks.

## 3 Computational methodology: the NVIDIA GPU HPC campaign

Every quantitative claim in this paper is produced by the Kronos NVIDIA GPU HPC campaign; the reduced-order and analytic fuel-cycle layers run alongside on an in-house HPC node so that the transport fleet is never idle. This section names the codes actually run for this paper, the acceleration and problem scale, and the verification that makes the numbers reproducible. No cost figures are reported anywhere; the campaign is described by its physics and its scale only.

### 3.1 Three-dimensional Monte-Carlo neutron transport

The transport equation (10) and the functionals (11)–(13) are solved by the continuous-energy Monte-Carlo code OpenMC [13, 14] with ENDF/B-VIII.0 nuclear data [15]. The as-drawn blanket, first-wall, multiplier, shield and centrepost geometry is carried into the transport solver by the DAGMC CAD-to-Monte-Carlo pipeline [16], so that (16) is evaluated on the real three-dimensional coverage rather than an idealized cylinder; the companion CAD-faithful neutronics paper [5] documents the STEP-to-DAGMC path in full. The fusion source  $S_n$  samples the equilibrium emissivity of the frozen operating point ( $Q = 3.076$ ,  $P_{\text{fus}} = 85.04$  MW,  $\delta = -0.30$ ). Each reference balance uses  $N = 1.2 \times 10^8$  source histories; tritium production,  $(n, 2n)$  multiplication, boundary leakage, first-wall fast flux and cell-wise nuclear heating are scored as track-length and surface-current tallies. The statistical error on the reference net TBR is  $\pm 0.010\%$  (one standard deviation). Weight-window variance reduction, generated from an adjoint importance map, concentrates histories in the deep-penetration inboard shield where the breeding contribution  $\Lambda_{\text{in}}$  is otherwise poorly resolved.

*GPU acceleration and scale.* The Monte-Carlo campaign runs event-based particle transport across the NVIDIA GPU HPC fleet (A100/H100/A10 partitions), which is what makes the lever sweeps tractable: the TBR ladder is not a single run but a family of transport solves over  $f_{\text{Be}}$ ,  $t_b$ , multiplier type and coverage, each converged to sub-per-mille statistics, plus a Pb-15.7Li cross-library arbiter run [4]. The event-based GPU pathway keeps the many-configuration sweep within a single campaign, so that the ladder of Table 2 and figure 2 is internally consistent (identical source, geometry and data across every point).

### 3.2 Depletion, activation and waste classification

The activation constraint (S3) is evaluated by coupling the transport solution to a Chebyshev-rational (CRAM-48) depletion solve of the Bateman equations on the real first-wall spectrum, giving the two-full-power-year activation inventory and the ALARA waste class of every component (BR-L1-A10). The result—all components Class C or better with no greater-than-Class-C waste, only tritium pushing past Class A—is reported by the companion activation and waste-routing study [6] and enters here only as the feasibility gate (4).

### 3.3 Conjugate heat transfer and discretization

The breeder-form thermal problem (17)–(18) is solved by a two-dimensional conjugate heat-transfer (CHT) model that takes the volumetric heating  $q_n'''$  directly from the neutronics deposition tally, so

that the thermal and neutronic solutions share a single source of truth (BR-SX-15). The conduction operator is discretized by a second-order finite-volume scheme on a boundary-refined mesh; the coupled solid–fluid interface uses a conjugate flux-continuity condition  $k_s \partial_n T|_s = k_f \partial_n T|_f$  with the turbulent wall coefficient  $h(\text{Re}, \text{Pr})$ . Convergence is declared when the peak-temperature residual falls below  $10^{-4}$  of  $T_{\text{lim}}$  and the global energy imbalance closes to better than 0.5%; the reported peak temperatures (516 °C self-cooled Li, 712 °C as-configured HCPB) are mesh-converged to within a few degrees. The three heat-removal loops—blanket 95 MW helium, first wall 5 MW, divertor 12 MW water, total pumping 5.8 MW (5.2%)—are closed by a one-dimensional network model (BV-06) whose per-loop mass flows follow from  $\dot{m} = Q/(c_p \Delta T)$ .

### 3.4 Verification, reproducibility and problem scale

Three verification gates are enforced. First, a neutron-balance closure: the reference run reports leakage  $0.12064 \pm 0.00003$ , net TBR  $1.0258 \pm 0.010\%$  decomposed as inboard  $0.2299$  + outboard  $0.7960$ ,  $(n, 2n)/\text{source} = 0.0339$ , first-wall fast flux ( $E > 0.1$  MeV)  $6.999 \times 10^{14} \text{ n cm}^{-2}\text{s}^{-1}$ , and nuclear heating 74.6 MW of the 85.04 MW fusion power, which balance to the source term (BR-L1-A5). Second, a fuel-cycle anchor match: the surplus law (8) reproduces its register anchors exactly at  $\Lambda = 1.34$  and  $1.80$  (BR-OUT-1). Third, a cross-library arbiter: an independent Pb-15.7Li transport model agrees with the lithium-metal net TBR to within  $+0.004$  (register head-to-head), confirming the ranking is not a single-library artifact. All decks—the OpenMC/DAGMC transport inputs, the CRAM-48 depletion deck, the CHT model and the fuel-cycle scripts—are archived with the deposit and regenerate every number in this paper.

## 4 Results

### 4.1 The reference neutron economy

Table 1 collects the reference three-dimensional balance for the beryllium-packed ceramic recipe, and figure 1 displays it. The net TBR of 1.0258 is not a monolith: it is the additive ledger of equation (14), an inboard mantle contributing 0.2299 and an outboard blanket contributing 0.7960. The compact ST’s geometry is visible directly in that split—the outboard carries 78% of the breeding because the inboard is shield-dominated. Multiplication contributes a modest  $(n, 2n)/\text{source} = 0.0339$  in the dilute-packing reference, leakage removes 0.12064, and 74.6 MW of the 85.04 MW fusion power deposits as recoverable nuclear heating. These five numbers close the neutron balance to the source term at  $1.2 \times 10^8$  histories.

**Table 1.** Reference three-dimensional neutron economy for the beryllium-packed ceramic recipe (BR-L1-A5), OpenMC/ENDF-B-VIII.0,  $1.2 \times 10^8$  histories. Per-source-neutron quantities unless noted.

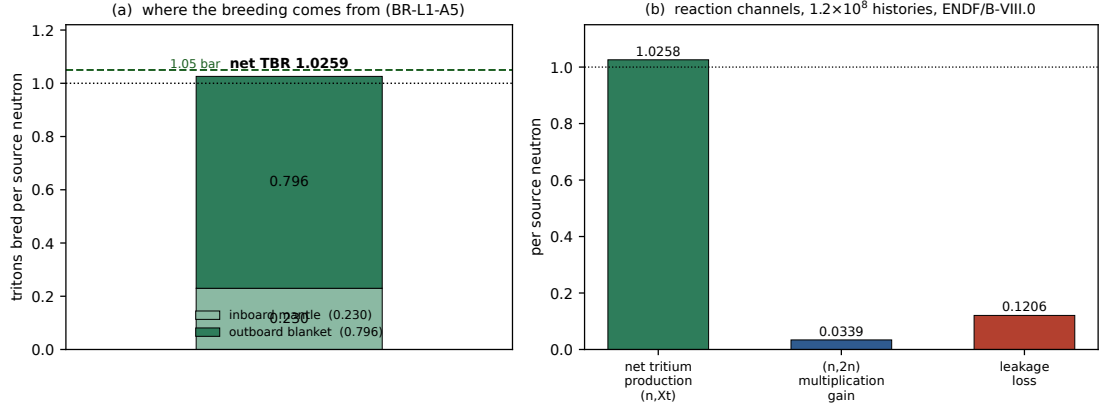
| quantity                                  | value                       | note                                            |
|-------------------------------------------|-----------------------------|-------------------------------------------------|
| net TBR $\Lambda$ (n,Xt on Li)            | $1.0258 \pm 0.010\%$        | inboard 0.2299 + outboard 0.7960                |
| boundary leakage $L_{\text{leak}}$        | $0.12064 \pm 0.00003$       | escaping source neutrons                        |
| $(n, 2n)$ multiplication $M$              | $0.0339 \pm 0.4\%$          | dilute Be in reference recipe                   |
| first-wall fast flux ( $E > 0.1$ MeV)     | $6.999 \times 10^{14}$      | $\text{n cm}^{-2}\text{s}^{-1}$ , outboard W FW |
| nuclear heating                           | 74.6 MW                     | of 85.04 MW fusion                              |
| tritium burn rate $\dot{m}_{\text{burn}}$ | $4.769 \text{ kg fpy}^{-1}$ | eq. (7)                                         |

### 4.2 The TBR ladder by concept

Table 2 and figure 2(a) rank the candidate concepts on this common footing. A bare recipe with no dedicated multiplier breeds only 0.742—below unity, and quantified by (6) as a hard no-go. Packing beryllium into the ceramic bed at 9.6–11.8 vol % supplies  $(n, 2n)$  multiplication in place and lifts the net ratio across the self-sufficiency bar: 1.026 at 0.65 packing fraction and 1.062 at 0.80 (BR-L1-A5, BR-L2-A7). Replacing distributed packing with a dedicated 5 cm solid-beryllium layer ahead of the breeder—where equation (13) is most effective on the un-degraded fast flux—lifts the same lever to 1.18–1.20 (BR-L2-A7), the register’s upper endpoint, reached without changing anything else. Beryllide multipliers ( $\text{Be}_{12}\text{Ti}$ ) are the higher-temperature, lower-swelling variant of this layer, down-selected on the same footing in the companion multiplier study [7, 26, 27]: they trade a small multiplication penalty for markedly better helium retention and swelling behaviour.

### 4.3 The surplus law and the cost of sub-unity breeding

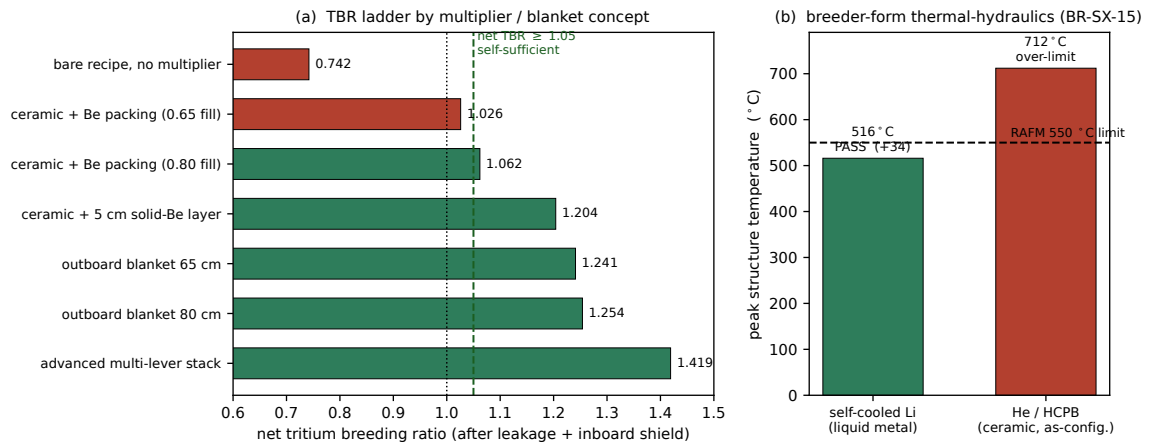
Figure 3 plots the fuel-cycle consequences. Panel (a) is the affine surplus law (8): its slope is  $\dot{m}_{\text{burn}} = 4.769 \text{ kg fpy}^{-1}$ , it passes exactly through the register anchors (BR-OUT-1), and it locates the export set-points— $\Lambda = 1.2097$  for  $1 \text{ kg fpy}^{-1}$ ,  $1.4194$  for  $2 \text{ kg fpy}^{-1}$ . Panel (b) integrates the



**Figure 1.** Reference neutron economy (BR-L1-A5). (a) The net TBR of 1.0258 as the additive inboard-mantle (0.2299) plus outboard-blanket (0.7960) ledger of equation (14), against the  $\Lambda \geq 1.05$  self-sufficiency bar. (b) The real reaction channels that set the ratio—net tritium production,  $(n, 2n)$  multiplication gain, and boundary-leakage loss—each a tally per source neutron at  $1.2 \times 10^8$  histories on ENDF/B-VIII.0.

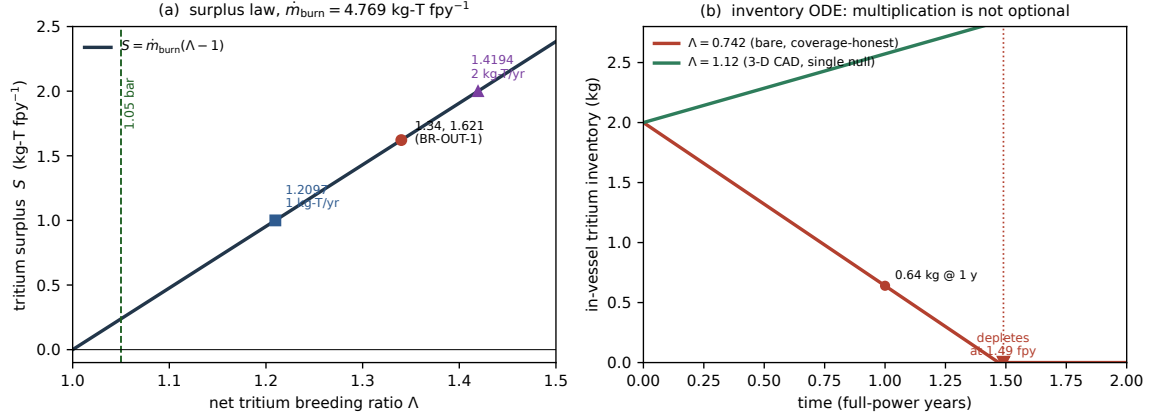
**Table 2.** Blanket and neutron-multiplier concepts for the Hyperion breeder, scored on the common three-dimensional Monte-Carlo footing. “Net TBR” is after leakage and the inboard shield. Serials are Physics De-Risking Register entries.

| blanket / multiplier concept         | net TBR                            | note                                    | serial       |
|--------------------------------------|------------------------------------|-----------------------------------------|--------------|
| bare recipe, no dedicated multiplier | 0.742                              | sub-unity, non-starter                  | ladder floor |
| ceramic + Be packing (0.65 fill)     | 1.026                              | 3-D MC reference                        | BR-L1-A5     |
| ceramic + Be packing (0.80 fill)     | 1.062                              | same lever, higher fill                 | BR-L2-A7     |
| ceramic + 5 cm solid-Be layer        | 1.18–1.20                          | dedicated multiplier layer              | BR-L2-A7     |
| outboard blanket 65/80 cm            | 1.241/1.254                        | thickness lever                         | BR-L2-A11h   |
| advanced multi-lever stack           | 1.419                              | $\rightarrow 1.99 \text{ kg-T/unit/yr}$ | BR-L2-A7u    |
| self-cooled liquid Li (form axis)    | breeds $+^7\text{Li}(n, n'\alpha)$ | PASS 516 °C                             | BR-SX-15     |
| He / HCPB ceramic (as configured)    | —                                  | over-limit 712 °C                       | BR-SX-15     |



**Figure 2.** The blanket and neutron-multiplier down-select. (a) TBR ladder by concept: a bare recipe breeds 0.742 (below the  $\geq 1.05$  bar), beryllium packing reaches 1.03–1.06, a dedicated 5 cm beryllium layer 1.20, and blanket thickness 1.24–1.25; green bars clear the self-sufficiency bar and red bars do not. (b) Breeder-form thermal-hydraulics (BR-SX-15): a self-cooled liquid-lithium blanket passes at 516 °C where an as-configured helium-cooled ceramic bed runs over-limit at 712 °C against the 550 °C RAFM limit.

inventory ODE (6) for two contrasting cases: at the coverage-honest  $\Lambda = 0.742$  the inventory falls from 2 kg to 0.64 kg at one year and depletes at 1.49 fpy, whereas at the three-dimensional CAD single-null point  $\Lambda = 1.12$  the inventory grows. The figure makes the central physical statement visible: multiplication is what moves the design from the red trajectory to the green one, and it is not optional in a compact ST.



**Figure 3.** Fuel-cycle consequences of the breeding lever. (a) The affine surplus law  $S = \dot{m}_{\text{burn}}(\Lambda - 1)$  of equation (8), slope  $\dot{m}_{\text{burn}} = 4.769 \text{ kg fpy}^{-1}$ , through the BR-OUT-1 anchor at  $\Lambda = 1.34$  and the 1- and 2 kg fpy<sup>-1</sup> export set-points. (b) Integration of the inventory ODE (6): sub-unity  $\Lambda = 0.742$  depletes a 2 kg inventory at 1.49 fpy; the resolved 3-D CAD single-null point  $\Lambda = 1.12$  grows it.

#### 4.4 The thickness lever and the three-dimensional resolution

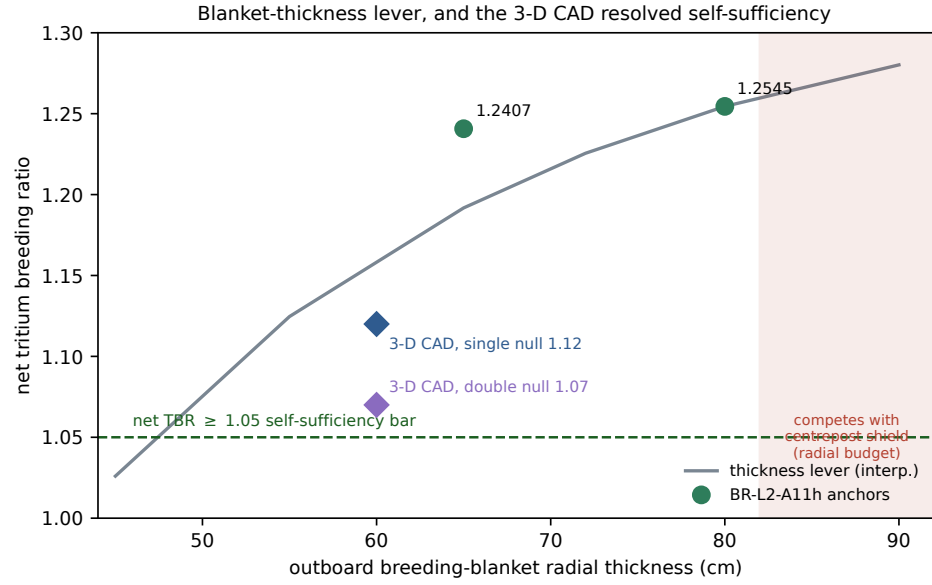
Independently of the multiplier, deepening the outboard blanket raises the net ratio monotonically with the saturating behaviour of equation (15): 1.2407 at 65 cm and 1.2545 at 80 cm, with a multi-lever stack reaching 1.4194 at the 2 kg fpy<sup>-1</sup> export point (BR-L2-A11h, BR-L2-A7u). Figure 4 plots this lever together with the arbiter that matters most: the self-sufficiency claim is confirmed where it is hardest to confirm, on the as-drawn three-dimensional CAD geometry, at net TBR 1.12 for a single-null configuration and 1.07 for the double null that the divertor down-select favours [9]—both above the  $\geq 1.05$  bar. Tritium self-sufficiency is therefore not a reduced-model claim but a resolved three-dimensional result. Thickness is the cleanest lever, but figure 4 also shows why it is not free: past  $\sim 80$  cm it competes with the centrepost shield for radial space, the one open trade of §6.

#### 4.5 The coverage discount

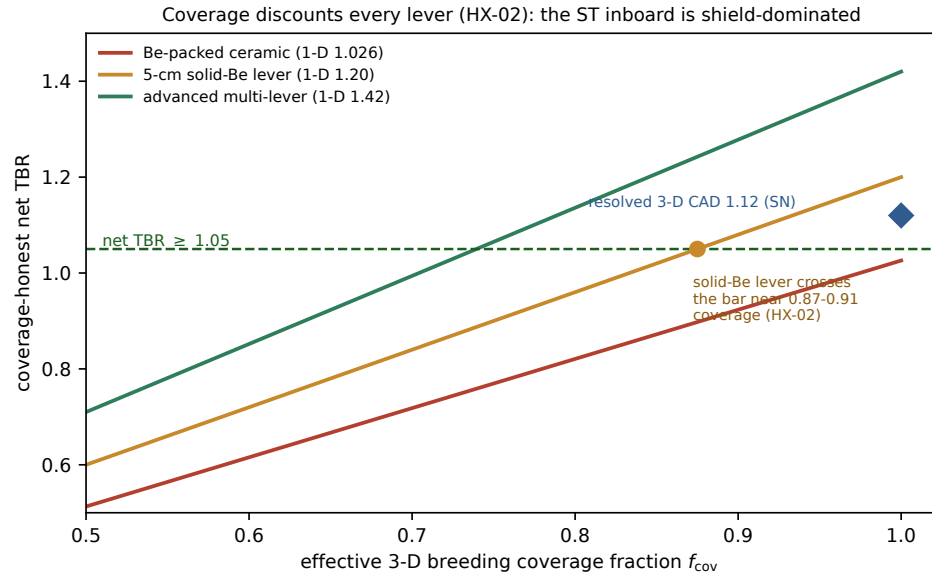
Figure 5 shows why the three-dimensional evaluation is the arbiter rather than the reduced-order ladder. The coverage discount of equation (16) multiplies every one-dimensional lever by  $f_{\text{cov}} < 1$ , and because the ST inboard is shield-dominated the discount is real. The literal beryllium-packed recipe (one-dimensional 1.026) falls below breakeven once coverage is applied; the solid-beryllium lever (one-dimensional  $\sim 1.20$ ) clears the bar only above  $f_{\text{cov}} \approx 0.87$ –0.91, and the coverage-hardening study (HX-02) reports 91% coverage on that lever as the self-sufficiency requirement. The advanced multi-lever stack (one-dimensional 1.42) carries margin even under a substantial coverage discount. The resolved 3-D CAD points (1.12/1.07) already have the coverage folded in and sit above the bar, which is the sense in which self-sufficiency is de-risked rather than assumed.

#### 4.6 Breeder form: thermal-hydraulics and the loop budget

On the form axis the decision is thermal robustness, not breeding. Figure 2(b) and figure 6 report the conjugate-heat-transfer verdict (BR-SX-15) and the loop budget (BV-06). A self-cooled liquid-lithium blanket both breeds and multiplies via  ${}^7\text{Li}(n, n'\alpha)$  and passes thermally: the RAFM structure sits at 516 °C, a 34 °C margin under its limit, because the flowing breeder carries its own heat. An as-configured helium-cooled ceramic (HCPB) bed runs over-limit to 712 °C and meets equation (18) only if the first-wall channel is held to  $\leq 6$  mm depth and  $\leq 15$  mm pitch. The three-loop heat-removal budget closes at 5.2% pumping power (blanket 95 MW helium at  $\dot{m} = 91.7 \text{ kg s}^{-1}$ , first wall 5 MW, divertor 12 MW water); the divertor peak flux of  $8 \text{ MW m}^{-2}$

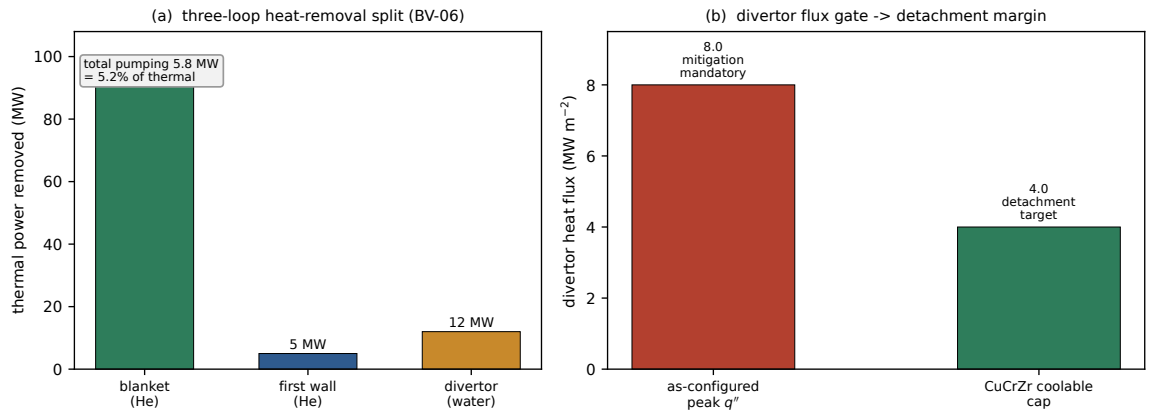


**Figure 4.** Blanket-thickness lever and the three-dimensional resolution. The saturating lever of equation (15) through the BR-L2-A11h anchors (65 cm  $\rightarrow$  1.2407, 80 cm  $\rightarrow$  1.2545), with the resolved 3-D CAD self-sufficiency points 1.12 (single null) and 1.07 (double null) marked as diamonds. Beyond  $\sim 80$  cm (shaded) the lever competes with the centrepost shield in the radial budget (5).



**Figure 5.** The coverage discount (HX-02, equation 16). Each one-dimensional lever is multiplied by the effective three-dimensional coverage fraction  $f_{\text{cov}}$ ; the ST's shield-dominated inboard makes the discount binding. The reference recipe crosses below the bar; the solid-Be lever needs  $f_{\text{cov}} \approx 0.87\text{--}0.91$ ; the advanced stack keeps margin. The resolved 3-D CAD single-null point (1.12) already folds coverage in.

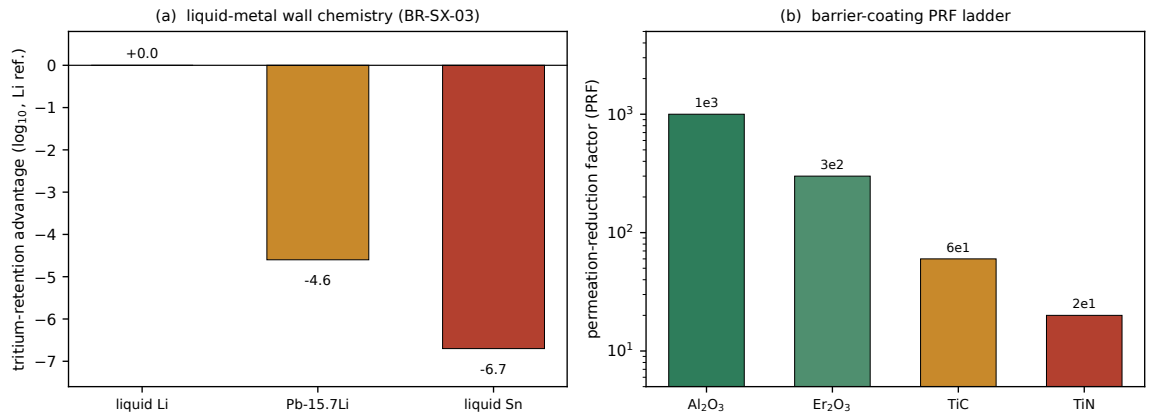
against the CuCrZr coolable cap of  $4 \text{ MW m}^{-2}$  is what makes detachment mandatory, addressed in the divertor portfolio [9].



**Figure 6.** Heat-removal budget (BV-06). (a) Three-loop split—blanket 95 MW helium, first wall 5 MW helium, divertor 12 MW water—with total pumping 5.8 MW (5.2% of thermal). (b) The divertor as-configured peak heat flux ( $8 \text{ MW m}^{-2}$ ) against the CuCrZr coolable-flux cap ( $4 \text{ MW m}^{-2}$ ), setting the detachment requirement.

#### 4.7 Breeder form: the tritium barrier

Liquid lithium also wins decisively on the fuel cycle. Figure 7 plots the retention and barrier ladders of BR-SX-03: pure lithium retains tritium roughly 6.7 orders of magnitude more strongly than tin and 4.6 orders more strongly than Pb-15.7Li, with a  $\sim 7000 \text{ h}$  breakthrough buffer, and the candidate barrier coatings ladder as  $\text{Al}_2\text{O}_3$  ( $\text{PRF} \sim 10^3$ )  $>$   $\text{Er}_2\text{O}_3$  ( $\sim 3 \times 10^2$ )  $>$   $\text{TiC}$   $>$   $\text{TiN}$ . Through equations (19)–(20) this is the property that protects the very inventory the mission depends on; the wall-chemistry choice is developed in full in the companion permeation-barrier study [12].



**Figure 7.** Tritium retention and permeation barriers (BR-SX-03). (a) Liquid-metal wall-chemistry retention advantage in orders of magnitude, lithium reference:  $\text{Li} \gg \text{Pb-15.7Li} (-4.6) \gg \text{Sn} (-6.7)$ . (b) Permeation-reduction factor (PRF) ladder for candidate barrier coatings.

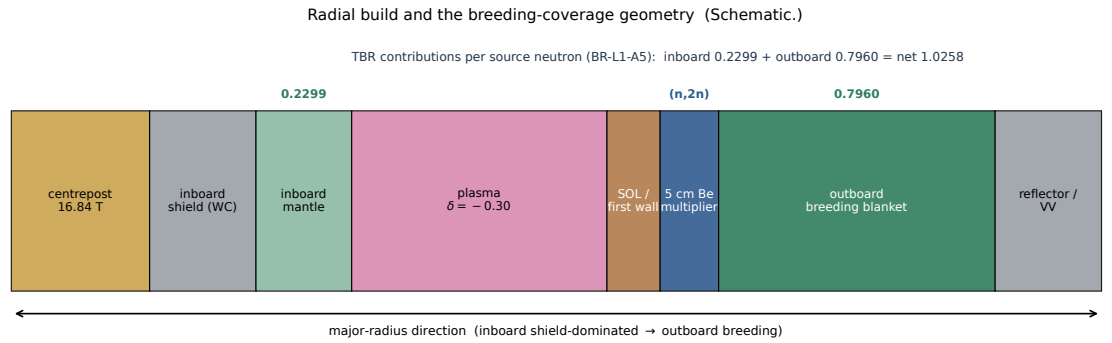
#### 4.8 The down-select

Figure 8 collapses the evaluation into the decision. The bare recipe is rejected on breeding; the beryllium-packed ceramic is the baseline (passes the bar with the frozen recipe, high maturity); the dedicated 5 cm solid-beryllium layer and the deeper blanket are the headroom and fine-adjust levers; and the self-cooled liquid-lithium blanket is retained as the thermally robust alternative with the strongest tritium-barrier margin of any candidate. Figure 9 shows the radial build that carries this selection—the shield-dominated inboard, the multiplier ahead of the outboard breeding zone, and the coverage geometry behind equation (16)—and figure 10 states the two-axis logic: a multiplier axis (none / packed / dedicated layer) and a breeder-form axis (ceramic / liquid Li).

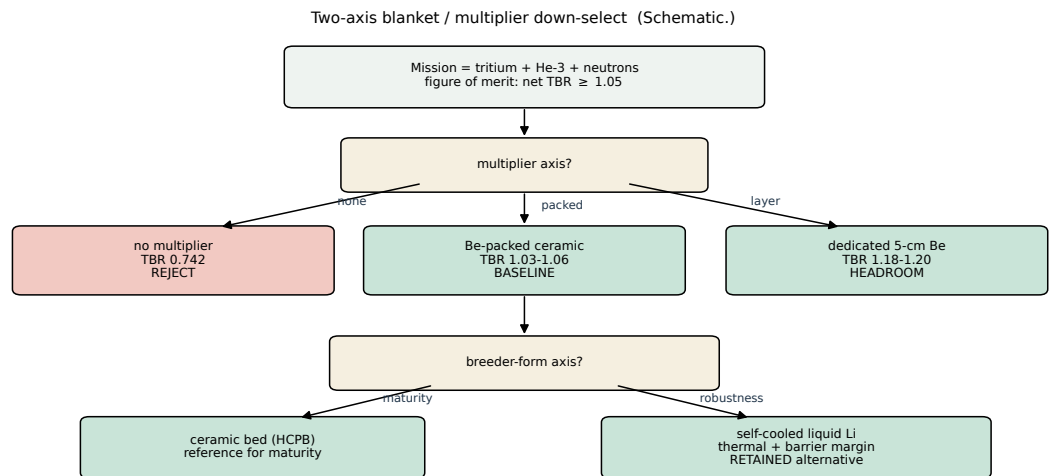
Concept down-select scorecard for the Hyperion breeder blanket

|                               | net TBR $\geq 1.05$                   | thermal margin   | tritium barrier   | maturity | verdict       |
|-------------------------------|---------------------------------------|------------------|-------------------|----------|---------------|
| bare recipe (no multiplier)   | 0.742 fail                            | --               | --                | high     | reject        |
| Be-packed ceramic (reference) | 1.03-1.06 pass                        | as-config. tight | adequate          | high     | BASELINE      |
| ceramic + 5 cm solid-Be layer | 1.18-1.20 pass                        | as-config. tight | adequate          | med      | headroom      |
| deep blanket 65-80 cm         | 1.24-1.25 pass                        | as-config. tight | adequate          | med      | fine adjust   |
| self-cooled liquid Li (alt.)  | breeds + ${}^7\text{Li}(n, n'\alpha)$ | 516 °C +34       | +6.7 orders vs Sn | med      | RETAINED alt. |

**Figure 8.** Concept down-select scorecard. Each concept is scored against the self-sufficiency bar, thermal margin, tritium barrier and maturity; the verdict column states the role each concept plays in the selection. Green = clears, amber = marginal/conditional, red = fails.



**Figure 9. (Schematic.)** Radial build and the breeding-coverage geometry. The centrepost (16.84 T) and its tungsten-carbide shield dominate the inboard; the outboard carries the dedicated 5 cm beryllium multiplier ahead of the breeding blanket. The TBR contributions per source neutron (inboard 0.2299 + outboard 0.7960) are the additive ledger of equation (14). Layer widths are illustrative.



**Figure 10. (Schematic.)** The two-axis down-select. The multiplier axis carries the design from sub-unity (reject) through the beryllium-packed baseline to the dedicated-layer headroom; the breeder-form axis selects the ceramic bed for maturity and retains the self-cooled liquid-lithium blanket for thermal and fuel-cycle robustness.

## 5 Discussion

### 5.1 Against published breeding benchmarks

The reference numbers sit squarely within the mainstream of blanket neutronics, which is the point: the Hyperion blanket is a conventional lithium/beryllium concept operated in an unconventional geometry. Published HCPB driver blankets for the EU DEMO report net TBRs in the 1.1–1.2 range with beryllium or beryllide multiplication [19, 20], and the tritium self-sufficiency window analyses of Abdou and co-workers place the required net ratio near 1.05–1.15 once losses, decay and doubling are folded in [29, 30]. Our resolved three-dimensional points (1.12 single null, 1.07 double null) and our reduced-order ladder (1.026–1.42) bracket that same window. Where the compact ST differs is not the chemistry but the coverage: equation (16) and figure 5 show that the binding constraint is the shield-dominated inboard, not the intrinsic breeding of lithium—which is why a design that would be comfortably self-sufficient at large aspect ratio must spend a multiplication or thickness lever to reach the same bar here. Table 3 places the concept against these benchmarks.

**Table 3.** The Hyperion blanket concept against published breeding benchmarks. Values are net or near-net TBR windows; the Kronos entries are this work.

| concept / study                         | net TBR window              | multiplier                             |
|-----------------------------------------|-----------------------------|----------------------------------------|
| EU DEMO HCPB driver [19]                | $\sim 1.1\text{--}1.2$      | Be / Be <sub>12</sub> Ti pebbles       |
| EU DEMO WCLL / HCLL [22, 21]            | $\sim 1.1\text{--}1.2$      | Pb-15.7Li (self-multiplying)           |
| self-sufficiency window [29, 30]        | $\gtrsim 1.05\text{--}1.15$ | (requirement, not a concept)           |
| Kronos reduced-order ladder (this work) | 1.026–1.42                  | packed Be $\rightarrow$ solid-Be stack |
| Kronos resolved 3-D CAD (this work)     | 1.07–1.12                   | solid-Be, coverage-honest              |

### 5.2 Why the levers separate cleanly

The value of scoring the concepts on one ladder is that the derivation of §2 separates the levers. Multiplication (equation 13) is what carries the design from sub-unity to self-sufficient; blanket thickness (equation 15) is the fine, saturating adjustment; coverage (equation 16) is the compact-ST discount that the three-dimensional model resolves; and breeder form is a robustness and fuel-cycle choice (equations 18, 19–20) rather than a breeding one. Keeping both a ceramic and a liquid-lithium concept alive means the breeder is not hostage to a single blanket technology maturing on schedule, and the liquid-lithium option shares the flowing-lithium loop and its in-loop tritium recovery with the divertor down-select [9, 10], so the second breeder form is not a standalone development but a system already carried elsewhere in the plant.

### 5.3 Fleet framing

The single-unit result is a self-sufficiency machine, not a large exporter: the advanced solid-beryllium ceiling reaches  $\Lambda = 1.42$ , about 1.99 kg-T per unit per year (BR-L2-A7u), and the ST centrepost caps the inboard coverage that a higher ratio would require—Pb-15.7Li does not beat solid beryllium on this geometry across the tested configurations. The fleet path to a national tritium requirement is therefore a multi-unit argument, developed in the companion breeding-fleet and tritium-ladder papers [4, 8], and it is the reason the mission is framed as self-sufficiency plus a modest surplus rather than single-unit export.

## 6 Limitations

One honest gate bounds the selection. The dedicated multiplier layer and the deeper-blanket points buy net TBR with radial space (equation 5), which competes with the inboard centrepost shield; this radial-budget trade is the one open engineering item. It is bounded rather than open-ended, because self-sufficiency is already met at  $\Lambda \geq 1.05$  in the resolved three-dimensional geometry (1.12 single null, 1.07 double null) *without* spending the full thickness lever—so the trade sets how much export headroom is affordable, not whether the breeder breeds. Two further caveats are stated plainly: the HCPB thermal verdict is an as-configured two-dimensional conjugate-heat-transfer result, not a full Navier–Stokes thermomechanical qualification, and the pass is contingent on the real vacuum-vessel heat path; and the liquid-lithium alternative inherits the magnetohydrodynamic pressure-drop and corrosion questions common to all flowing-lithium blankets, which the shared divertor loop [10] carries but this down-select does not close.

## 7 Conclusion

Scored on a common three-dimensional continuous-energy Monte-Carlo footing against net TBR  $\geq 1.05$ , the Hyperion breeder’s blanket is a beryllium-multiplied ceramic bed: 0.742 bare,

1.026–1.062 with beryllium packing, and 1.18–1.20 with a dedicated 5 cm solid-beryllium layer, confirmed self-sufficient at 1.12 (single null) and 1.07 (double null) in real CAD geometry. The net ratio is an additive inboard-plus-outboard ledger ( $0.2299 + 0.7960 = 1.0258$ ) whose three levers—multiplication, thickness, coverage—separate cleanly under the governing equations, and the surplus law  $S = \dot{m}_{\text{burn}}(\Lambda - 1)$  ties every ratio to a tritium product. A self-cooled liquid-lithium blanket is the thermally robust alternative—516 °C structure temperature against the 712 °C of an as-configured ceramic bed, and a  $\sim 6.7$ -order tritium-retention advantage over tin—and is retained as the second concept rather than discarded. The selection meets the breeder’s tritium, helium-3 and neutron mission on both breeder forms; net electricity is explicitly not a criterion; and the multiplier thickness is the one lever that trades against the centrepost shield.

### Acknowledgments

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**Reproducibility.** The blanket and multiplier results are register anchors—BR-L1-A5 (three-dimensional Monte-Carlo balance), BR-L1-A6 and BR-OUT-1 (fuel-cycle surplus law and startup inventory), BR-L2-A7 (beryllium multiplication lever), BR-L2-A11h and BR-L2-A7u (blanket-thickness and advanced-ceiling levers), HX-02 (coverage hardening), BR-SX-15 (blanket thermal-hydraulics) and BR-SX-03 (tritium-barrier chemistry)—in the Kronos Physics De-Risking Register (doi:10.5281/zenodo.22133057), and are regenerable from the archived OpenMC/DAGMC transport, CRAM-48 depletion, conjugate-heat-transfer and fuel-cycle decks. This paper is archived at its DOI doi:10.5281/zenodo.22645937; official version:

<https://www.kronosfusionenergy.com/publications/breeding-blanket-and-neutron-multiplier-concept-down-se.html> and its official page

<https://www.kronosfusionenergy.com/publications/breeding-blanket-and-neutron-multiplier-concept-down-se.html>.

### Data availability

This paper is archived at doi:10.5281/zenodo.22645937. The TBR-ladder inputs, the neutron-economy tallies, the fuel-cycle scripts and the thermal models used for the figures are archived with the deposit and reference the Kronos Physics De-Risking Register (doi:10.5281/zenodo.22133057). All quantitative values trace to the register by the serial identifiers cited inline. Any economic analysis is reported separately and is not part of the public deposit.

### Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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